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Return to Play and Recurrence After Calf Muscle Strain Injuries in Elite Australian Football Players

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Background: Calf muscle strain injuries (CMSI) are prevalent in sport, but information about factors associated with time to return to play (RTP) and recurrence is limited.

Purpose: To determine whether clinical and magnetic resonance imaging (MRI) data are associated with RTP and recurrence after CMSI.

Study Design: Case-control study; Level of evidence, 3.

Methods: Data of 149 CMSI reported to the Soft Tissue injury Registry of the Australian Football League were explored to evaluate the impact of clinical data and index injury MRI findings on RTP and recurrence. Clinical data included age, previous injury history, ethnicity, and the mechanism of injury.

Results: Irrespective of the anatomical location, players with CMSI with severe aponeurotic disruption (AD) took longer to RTP than players with CMSI with no AD: 31.3 ± 12.6 days vs 19.4 ± 10.8 days (mean ± SD; $P = .003$). A running-related mechanism of injury was associated with a longer RTP period for CMSI overall (adjusted hazard ratio [AHR], 0.59; $P = .02$). The presence of AD was associated with a longer RTP period for soleus injuries (AHR, 0.6; $P = .025$). Early recurrence (ie, ≤2 months of the index injury) was associated with older age (AHR, 1.3; $P = .001$) and a history of ankle injury (AHR, 3.9; $P = .032$). Older age (AHR, 1.1; $P = .013$) and a history of CMSI (AHR, 6.7; $P = .002$) increased the risk of recurrence within 2 seasons. The index injury MRI findings were not associated with risk of recurrence.

Conclusion: A running-related mechanism of injury and the presence of AD on MRI were associated with a longer RTP period. Clinical rather than MRI data best indicate the risk of recurrent CMSI.

Keywords: calf injury; muscle strain; MRI; soleus; gastrocnemius; return to play

Calf muscle strain injuries (CMSI) are costly in elite sport, with a season prevalence of up to 6% in soccer[13] and 10.6 matches missed per club per season in Australian Football.[2] The reported return to play (RTP) periods for CMSI vary from no time loss to >120 days in American football,[13,18,56,57] and athletes can be prone to recurrence with rates as high as 14.4% in soccer[14] and 16% in Australian Football.[2] Despite the impact of CMSI, there is limited information about the factors that might be associated with outcomes after injury.

Clinical and magnetic resonance imaging (MRI) characteristics could affect the time to RTP. For example, time to RTP has been found to be 11.9 days longer for running-related CMSI as compared with CMSI resulting from non–running related activities.[18] The exact value of MRI in estimating time to RTP after CMSI is unclear. Injuries involving the central intramuscular aponeurosis of soleus (ie, central tendon)[4,7,22] have the longest reported average time to RTP (mean ± SD, 44.3 ± 23.0 days), but less is

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known about CMSI at other anatomical locations.[4,40,42,44] Evidence of aponeurotic disruption (AD) could also affect time to RTP.[3,9,44]

An understanding of the factors associated with recurrent CMSI is important, particularly considering recurrences are often afforded a longer rehabilitation period relative to the index injury.[14,18] MRI findings of the index injury might inform the risk of recurrence, along with other factors such as clinical data, tissue healing time frames (eg, aponeurotic vs epimysial tissue), and the demands of the sport.[37,38] Recurrent CMSI are common during rehabilitation or shortly after RTP, tend to afflict older athletes, and usually involve a running-related mechanism of injury.[18] It is unknown whether clinical and MRI data are associated with recurrence after CMSI.

The aims of this study were to (1) describe the MRI findings (including index vs recurrent injuries) and functional progression after CMSI occurring at various anatomical locations and (2) determine if clinical and MRI data concerning index CMSI are associated with time to RTP and recurrence. We hypothesized that running-related CMSI and soleus central intramuscular aponeurosis injuries showing evidence of AD would have the longest RTP periods. We also hypothesized that older age and a running-related mechanism of injury would be associated with a greater risk of recurrent CMSI.

METHODS

Study Procedure and Cohort

Clinical and MRI data were collected as part of the Soft Tissue injury Registry of the Australian Football League (AFL) over 4 consecutive seasons (2014-2017). Club medical personnel (doctor and/or physical therapist) recorded clinical data about injuries using a standard form for CMSI requiring modification and/or time loss from training or competition. Clinical data have been previously described,[18] and comprise the following:

- Player demographics, including the injury history at the time of the index injury (CMSI, ankle, foot)
- The training and match history prior to injury
- Injury circumstances, including the mechanism of injury: running-related mechanisms (acceleration, high intensity running, steady-state running, deceleration, sudden change of direction) and non–running related mechanisms (jumping, landing, kicking [stance leg], gradual onset, nil-specific, other)
- Days to achieve pain-free walking, running at >90% of maximum speed, return to full training, and RTP

All cases (N = 149) with MRI evidence of CMSI were included. Ethics approval was obtained from La Trobe University (HEC13-057) and the AFL Research Board. All players provided written consent.

Data Synthesis

Classification of Index and Recurrent CMSI. Injuries were classified as an index or a recurrent injury. A recurrence was defined as an MRI-confirmed subsequent CMSI that involved the same leg as the index injury. All injuries were followed for a maximum period of 2 seasons to determine if there was a recurrence.[15]

MRI Classification. Two blinded radiologists (M.L., A.H.R.) independently reviewed 149 deidentified MRI scans. Standardized classification and reporting included the following:

- Muscle(s) injured
- Gross injury position: superior-inferior, medial-lateral, and depth (superficial or deep)
- Anatomical location of injury: soleus (central, lateral, and medial intramuscular aponeuroses; posterior intermuscular aponeurosis; deep anterior aponeurosis; myofascial) and gastrocnemius (anterior intermuscular aponeuroses of the medial head and lateral head, proximal musculotendinous junction, distal musculotendinous junction, myofascial)[4,40,42,44,56,57]
- Presence of AD: yes/no
- Severity of AD: none (0%), mild (<50%), severe (between 50%-99%), complete (100%) of the structural cross-sectional area[41,42,51,52]
- Waviness: yes/no[28,47,51,52]
- Edema length: craniocaudal (in millimeters)
- Scar tissue (ie, from a previous injury): yes/no

If >1 muscle was injured, the muscle with the largest injury and/or severity of AD was classified as the primary muscle involved. If there was >1 anatomical location of injury, the site with the largest injury and/or severity of AD was classified as the primary anatomical location.

Outcome Measures. The number of days from the date of the injury to return to competition quantified the length of the RTP period. Functional progression during the RTP period was measured by the number of days to achieve pain-free walking, running at >90% of maximum speed, return to full training, and RTP. Recurrence was quantified as the cumulative incidence relative to the date of the index injury. Early recurrences were defined as occurring within 2 months from the date of the index injury.[36,55]

Statistical Analysis

MRI Characteristics. Frequency distributions were used to describe the prevalence of MRI findings:

- Gross injury position: superior-inferior, medial-lateral, superficial-deep
- Primary anatomical location of injury: soleus (central, lateral, and medial intramuscular aponeuroses; posterior intermuscular aponeurosis; myofascial) and gastrocnemius

(anterior intermuscular aponeurosis of the medial head, anterior intermuscular aponeurosis of the lateral head, myofascial)
- AD: presence and severity
- Waviness: visible connective tissue irregularity believed to be due to loss of structural integrity
- Scar tissue

The mean craniocaudal length of edema was calculated for CMSI at each anatomical location. The Fisher exact test was used to evaluate the association between the mechanism of injury and AD.

Anatomical Location of Injury and Functional Progression. Data were not normally distributed. The length of the RTP period was compared among anatomical locations for all CMSI (Kruskal-Wallis H test). The time to achieve pain-free walking, running at >90% of maximum speed, return to full training, and RTP were compared between CMSI involving the same anatomical location with and without AD (Mann-Whitney U test). Effect size (r) was calculated using: $r = z/\sqrt{n}$ (small, $r \leq 0.1$; medium, $r = 0.30$; large, $r \geq 0.50$).[16,31]

Clinical and MRI Data Associated With RTP and Recurrence. Clinical and MRI data were identified from previous studies of muscle strain injuries, including CMSI. Clinical data included age, injury history, and ethnicity, as well as the mechanism of injury and stage of the season (see the Appendix, available in the online version of this article).[18,19,40] The MRI data included the primary muscle injured, presence and severity of AD, strain position, and edema length.[41-44,51,52,56] The time to RTP was also evaluated for its possible association with recurrence.[12] Univariate relationships were visualized and explored (Kaplan-Meier survival curves, log-rank test, and hazard ratios [HRs]). Variables that did not violate assumptions and showed a univariate association were included in multivariable Cox regressions to evaluate potential independent associations (adjusted HR [AHR]) with RTP and recurrence. Analyses were performed for all CMSI and for soleus injuries only; a separate analysis for gastrocnemius injuries was not performed due to the limited number of cases. For time to RTP, injured players who did not RTP prior to the end of the season were censored to account for time loss associated with all CMSI. Censoring concluded at the end of the season during which the injury occurred to ensure inflated RTP periods were not recorded.[10,11] SPSS version 25.0 (IBM) was used, with significance set at $P < .05$.

RESULTS

Injury Description

The 149 cases were made up of 114 index and 35 recurrent CMSI. A single muscle was involved in 107 CMSI, 41 involved multiple muscles (2 muscles, n = 30; 3 muscles, n = 11), and there was 1 Achilles tendon injury. The soleus or gastrocnemius was the primary muscle involved in 95.9% of cases (Appendix Table A1, available online).

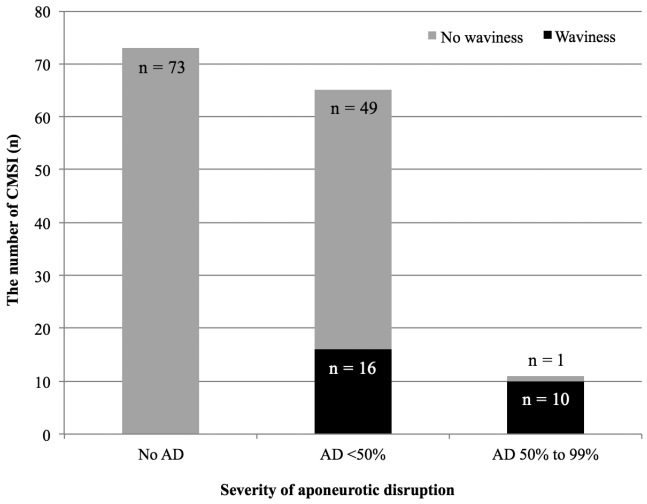

Figure 1. The prevalence of waviness with reference to the presence and severity of aponeurotic disruption (AD). CMSI, calf muscle strain injuries.

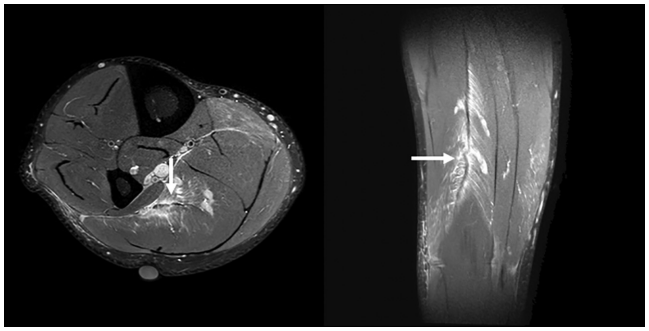

Figure 2. Soleus lateral intramuscular aponeurosis injury showing aponeurotic disruption (left image) and waviness (right image).

Most CMSI (68.5%) involved 1 anatomical location of injury. In 51% of CMSI the primary anatomical location of injury had AD. The severity was mild in 85.5% of these cases. There were no cases of complete AD. Waviness was present only in CMSI with AD. Almost all CMSI with severe AD had waviness (Figures 1 and 2).

Mechanism of Injury and Aponeurotic Disruption. Injuries without AD more likely had a nil-specific or gradual-onset mechanism of injury as compared with injuries with AD (52% vs 32.9%; odds ratio, 2.2; 95% CI, 1.1-4.3; $P = .013$). A running-related mechanism of injury was reported for 56.6% of injuries with AD and 39.7% of injuries with no AD (odds ratio, 1.8; 95% CI, 0.9-3.6; $P = .07$) (Figure 3).

Soleus. The soleus was the primary muscle involved in 94 index and 32 recurrent CMSI. Soleus injuries typically affected the middle third of the muscle and were deep (Table 1), with the most common anatomical locations being the lateral intramuscular aponeurosis (n = 35), medial intramuscular aponeurosis (n = 29), central

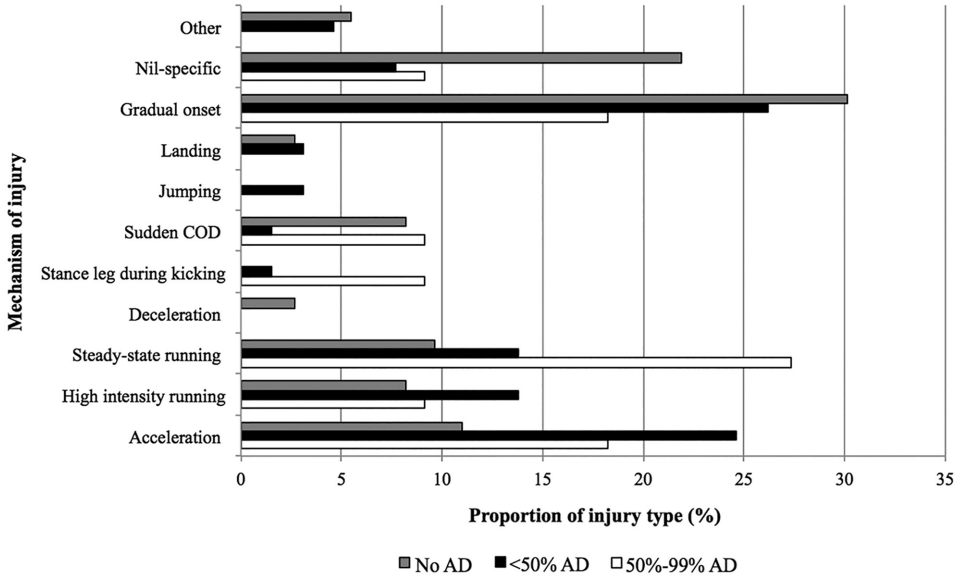

Figure 3. The mechanism of injury with reference to the presence and severity of aponeurotic disruption (AD). Proportions are expressed with reference to the category of AD. COD, change of direction.

TABLE 1
Primary Position and Anatomical Location of the 126 Soleus Injuries[a]

		Superior-Inferior, %				Medial-Lateral, %				Anterior-Posterior, %			
	%	Proximal	Middle	Distal	>1 Region	Medial	Middle	Lateral	>1 Region	Superficial	Deep	S&D	Edema Length,[b] Mean ± SD, mm
IMA	69.8	19.0	32.5	17.5	0.8	25.4	12	26.2	6.3	0.8	58.7	10.3	83.3 ± 37.8
Central	16.7	3.2	8.7	4.8	0.0	5.6	8.7	1.6	0.8	0.8	12.7	3.2	73.4 ± 28.9
Lateral	30.2	7.1	15.9	7.1	0.0	0.0	0.0	24.6	5.6	0.0	26.2	4.0	83.5 ± 37.2
Medial	23.0	8.7	7.9	5.6	0.8	19.8	3.2	0.0	0.0	0.0	19.8	3.2	90.1 ± 43.4
PA-S	15.9	1.6	6.3	7.9	0.0	10.3	0.0	5.6	0.0	15.1	0.0	0.8	89.9 ± 54.1
Myofascial	14.3	4.8	3.2	6.3	0.0	7.9	0.8	4.0	1.6	7.1	4.8	2.4	74.8 ± 46.8

[a]IMA, intramuscular aponeurosis; PA-S, posterior intermuscular aponeurosis of soleus; S&D, superficial and deep.
[b]Craniocaudal length of edema.

intramuscular aponeurosis (n = 21), and posterior intermuscular aponeurosis (n = 20). Only 11 (2.4%) soleus injuries affected the deep anterior aponeurosis. Half (50.8%) of the soleus injuries displayed AD (mild, 43.7%; severe, 7.1%). Eight of 9 soleus injuries with severe AD had waviness. The median time to recurrence was 58 days for soleus injuries. When compared with the index injury, recurrent soleus injuries primarily involved (1) the same muscle, gross position, and anatomical location (37.5%), or (2) the same muscle, but a different gross position and anatomical location (37.5%) (Figure 4A).

Gastrocnemius. The 14 index and 3 recurrent gastrocnemius injuries affected the distal (n = 8), middle (n = 5), and proximal (n = 4) regions. Edema length was on average 82.5 ± 50.8 mm for the 10 medial head injuries and 51.4 ± 26.7 mm for the 7 lateral head injuries. The anatomical location of injury involved the anterior intermuscular aponeurosis in 15 cases (medial head, 47.1%; lateral head, 41.1%). Nine gastrocnemius injuries had AD (mild, 41.2%; severe, 11.7%). The 2 cases with severe AD affected the medial head and showed waviness (Figure 5). The 3 recurrent gastrocnemius injuries all occurred early (median, 39 days); 2 of these involved the same muscle, gross position, and anatomical location as the index injury.

Return to Play

Anatomical Location of Injury and Functional Progression. The anatomical location of injury was unrelated to the length of the RTP period when considering all CMSI (χ^2 = 5.7; *df* = 6; *P* = .46) and when considering only CMSI with AD (χ^2 = 7.1; *df* = 5; *P* = .22) (Appendix Tables A3 and A4, available online). Soleus intramuscular aponeurosis injuries with AD had a slower functional progression when compared with injuries at the same location with no AD (Table 2). Patients with myofascial injuries had the shortest time on average to achieve pain-free

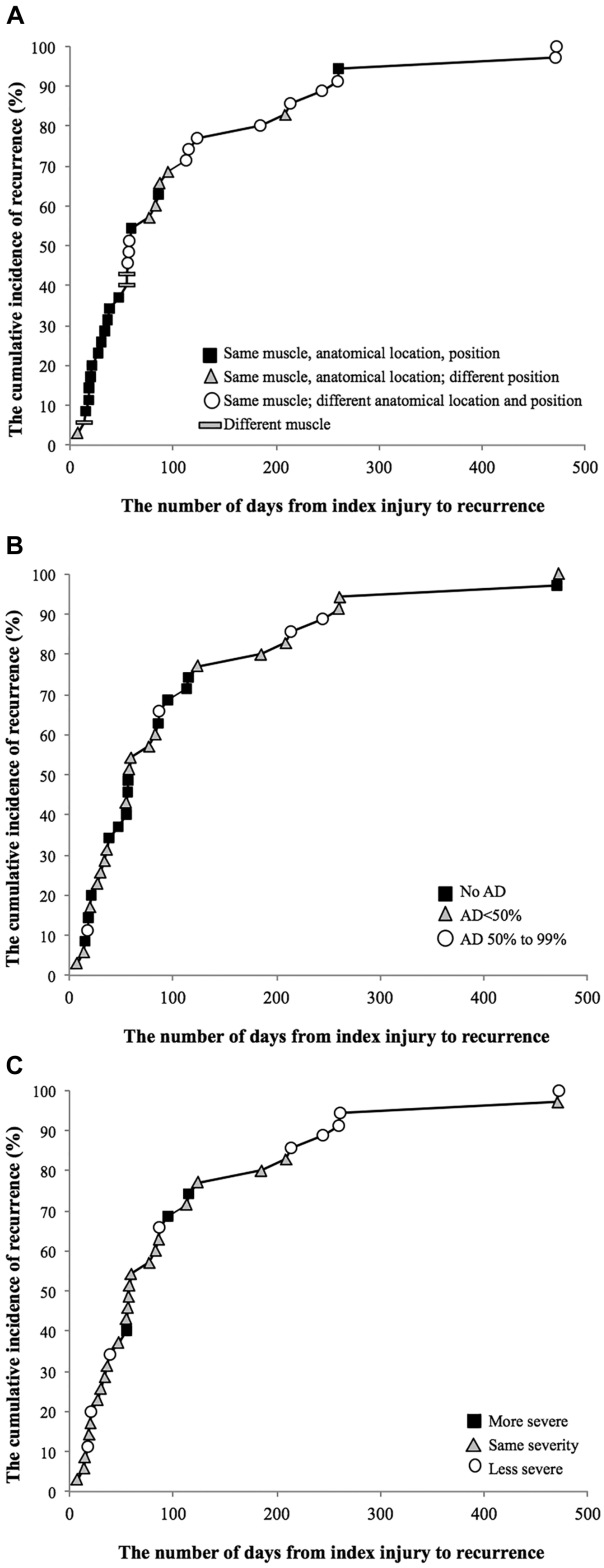

Figure 4. The cumulative incidence of recurrence: (A) the primary location of the recurrence vs the location of the index injury, (B) the presence and severity of aponeurotic disruption (AD) of the index injury, and (C) the severity of AD of the recurrence vs the index injury.

walking (3.1 ± 2.6 days), running at >90% of maximum speed (12.9 ± 5.4 days), return to full training (14.1 ± 5.2 days), and RTP (16 ± 5.6 days). Soleus intramuscular aponeurosis injuries with AD had significantly longer RTP periods than myofascial injuries (central intramuscular aponeurosis, $r = 0.58$, $P = .016$; lateral intramuscular aponeurosis, $r = 0.53$, $P = .007$; medial intramuscular aponeurosis, $r = 0.57$, $P = .010$).

Clinical and MRI Data Associated With RTP. Large variation was found in time to RTP (Figure 6). Injuries with no AD took on average 19.4 ± 10.8 days to RTP (median, 15; interquartile range [IQR], 13-23). Injuries with mild AD took 26.7 ± 16.8 days (median, 22.5; IQR, 14-37.5). Injuries with severe AD took 31.3 ± 12.6 days (median, 28; IQR, 23-41). There was no difference in time to RTP between CMSI with no AD versus mild AD ($r = 0.22$; $P = .06$), or mild AD versus severe AD ($r = 0.15$; $P = .30$). The RTP period was longer for CMSI with severe AD as compared with CMSI with no AD ($r = 0.37$; $P = .003$). Irrespective of severity, CMSI with AD took on average 8.4 days longer to RTP than CMSI with no AD (19.4 ± 10.8 vs 27.8 ± 15.9 days; $r = 0.27$; $P = .007$).

When evaluated together, clinical and MRI data were not associated with time to achieve pain-free walking (Appendix Tables A5 and A6, available online). Primary involvement of soleus (all CMSI: AHR, 0.59; $P = .042$) and a running-related mechanism of injury (all CMSI: AHR, 0.57; $P = .012$; soleus injuries only: AHR, 0.46; $P = .001$) were associated with longer time to achieve running at >90% of maximum speed (Table 3). A running-related mechanism of injury (AHR, 0.56; $P = .014$) and the presence of AD (AHR, 0.59; $P = .018$) were associated with a longer time to return to full training for soleus injuries. A running-related mechanism of injury was the only factor associated with a longer RTP period for CMSI overall (AHR, 0.59; $P = .019$), whereas the presence of AD (AHR, 0.6; $P = .025$) together with a running-related mechanism of injury (AHR, 0.55; $P = .01$) were associated with a longer RTP period for soleus injuries.

The soleus was the primary muscle involved in 32 recurrent CMSI (lateral intramuscular aponeurosis, 33.3%; medial intramuscular aponeurosis, 30%; posterior intermuscular aponeurosis, 20%). Most (63.2%) of the early recurrences affected the same muscle, gross position, and anatomical location as the index injury (Figure 4A), and the index injury involved AD (Figure 4B). Recurrent CMSI affecting the same muscle, anatomical location, and position had the same (71.4%) or less severe (28.6%) AD (Figure 4C). The index injury MRI findings (Appendix Tables A7 and A8, available online) and the time to RTP were not associated with the risk of recurrent CMSI (early: HR, 0.99; $P = .41$; overall: HR, 1.01; $P = .39$). Older age (all CMSI: AHR, 1.3; $P = .001$; soleus injuries only: AHR, 1.2; $P = .007$) and a history of ankle injury (all CMSI: AHR, 3.9; $P = .032$) were associated with a greater risk of CMSI recurring early. Older age (all CMSI: AHR, 1.1; $P = .013$) and a history of CMSI prior to the index injury (all CMSI: AHR, 6.7; $P = .002$; soleus injuries only: AHR, 5.6; $P = .005$) were associated with a greater risk of recurrent CMSI overall.

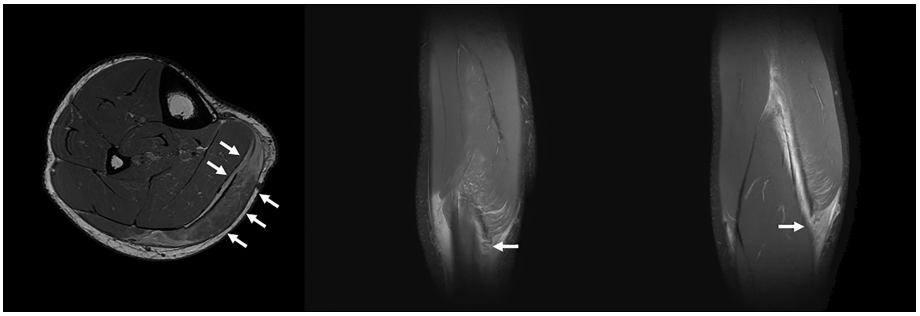

Figure 5. A medial gastrocnemius injury showing severe aponeurotic disruption (left image), waviness (middle image), and retraction (right image).

TABLE 2
Time to Achieve Pain-Free Walking, Running at >90% of Maximum Speed, Return to Full Training, and Return to Play for Each Anatomical Location of Injury With and Without Aponeurotic Disruption[a]

	Pain-Free Walking				Running at >90% of Maximum Speed				Return to Full Training				Return to Play			
	Mean ± SD	Range	95% CI	*r*	Mean ± SD	Range	95% CI	*r*	Mean ± SD	Range	95% CI	*r*	Mean ± SD	Range	95% CI	*r*
Central IMA																
No AD	4.2 ± 2.5	1-10	2.6-5.8		16.0 ± 8.9	6-38	9.9-22.0		18.5 ± 11.6	6-49	11.1-25.8		21.7 ± 12.2	7-49	13.9-29.4	
AD	6.0 ± 4.5	1-14	1.8-10.2	0.18	24.4 ± 12.9	10-40	12.5-36.4	0.32	26.3 ± 11.9	12-45	15.3-37.3	0.38	32.0 ± 13.9[b]	15-50	19.1-44.9	0.44
Lateral IMA																
No AD	3.1 ± 2.9	0-9	1.5-4.6		11.3 ± 5.1	2-25	8.7-13.9		11.8 ± -5.1	2-25	9.2-14.4		14.4 ± 5.3	4-28	11.6-17.1	
AD	4.3 ± 3.4	0-16	2.6-6.0	0.25	26.7 ± 16.2[b]	7-63	17.7-35.6	0.61	28.8 ± 13.0[b]	10-56	21.6-36.0	0.65	35.1 ± 16.9[b]	13-74	25.8-44.5	0.69
Medial IMA																
No AD	3.4 ± 2.3	0-8	2.3-4.5		15.4 ± 8.5	7-38	11.1-19.8		15.3 ± 7.8	7-35	11.3-19.3		19.9 ± 9.7	10-44	14.9-24.9	
AD	4.2 ± 1.9	1-7	2.9-5.4	0.20	19.6 ± 7.7	5-28	14.4-24.8	0.30	24.2 ± 11.9[b]	5-49	16.2-32.2	0.36	29.8 ± 9.7[b]	15-49	22.9-36.7	0.51
PA-S																
No AD	1.8 ± 2.0	0-4	0.7-4.3		10.8 ± 4.1	5-16	5.7-15.9		11.0 ± 4.5	7-18	5.4-16.6		16.6 ± 5.6	9-23	9.6-23.5	
AD	3.6 ± 2.9	0-9	2.1-4.9	0.25	18.9 ± 10.3	6-38	13.8-24.0	0.33	20.3 ± 11.9	6-45	14.4-26.2	0.35	25.0 ± 13.0	7-49	18.3-31.7	0.25
AA-MG																
No AD	2.0 ± 1.0	1-3	0.5-4.5		7.7 ± 7.4	2-16	10.6-25.9		9.0 ± 8.2	2-18	11.3-29.3		11.3 ± 10.1	2-22	3.6-36.3	
AD	6.2 ± 6.8	0-16	2.3-14.7	0.12	18.8 ± 15.8	5-44	0.8-38.4	0.10	22.6 ± 20.5	5-53	2.8-48.1	0.10	25.8 ± 22.7	6-58	2.4-53.9	0.12
AA-LG																
No AD	4.5 ± 4.0	1-10	1.9-10.9		12.5 ± 6.4	6-18	2.3-22.7		13.0 ± 7.3	7-23	1.3-24.7		17.0 ± 9.8	8-29	1.4-32.6	
AD	3.7 ± 3.5	0-7	1.0-12.4	0.13	17.3 ± 11.0	14-28	1.4-38.7	0.19	18.7 ± 8.1	10-30	1.2-44.7	0.20	19.7 ± 9.8	14-31	4.7-44.0	0.20

[a]Effect size: small, $r \leq 0.1$; medium, $r = 0.30$; large, $r \geq 0.50$. AD, aponeurotic disruption; AA-LG, anterior aponeurosis of lateral gastrocnemius; AA-MG, anterior aponeurosis of medial gastrocnemius; IMA, intramuscular aponeurosis; PA-S, posterior intermuscular aponeurosis of soleus.
[b]Statistically significant difference vs injuries at the same anatomical location that do not have aponeurotic disruption ($P < .05$).

DISCUSSION

This study reported 3 key findings about RTP and recurrence after CMSI in elite Australian Football players. First, the anatomical location of injury was not associated with time to RTP. Second, the mechanism of injury and MRI evidence of AD help inform the time taken to RTP. Third, clinical data about the athlete, such as age and injury history, may best evaluate the risk of recurrent CMSI. Index injury MRI findings do not. These findings advance our understanding of RTP and recurrence of CMSI beyond the broad epidemiological characteristics of these injuries provided in our previous study.[18]

We previously found functional progression and RTP periods to be similar for soleus versus gastrocnemius injuries.[18] Here, we have expanded on these initial findings and evaluated whether such outcomes were influenced by more specific injury characteristics observed on MRI. The anatomical location of injury was unrelated to the length of the RTP period. In contrast to Pedret et al,[40] we did not find soleus central intramuscular aponeurosis injuries to have the longest time to RTP. Different cohorts may explain the conflicting findings: elite-level Australian Football players versus a diverse group of athletes.[40] Our study also had 82 more soleus injuries. Consistent with previous studies, we found large variation in the RTP period for injuries at each anatomical location.[42,56,57] The MRI data evaluated in this study represent only some of the many variables that could influence the time to RTP after CMSI.

Evidence of AD (present: yes/no) was associated with a longer time to return to full training and RTP for soleus injuries. A prolonged RTP period may result because aponeuroses are associated with a slower rate of healing and adaptation upon re-loading due to their structure and

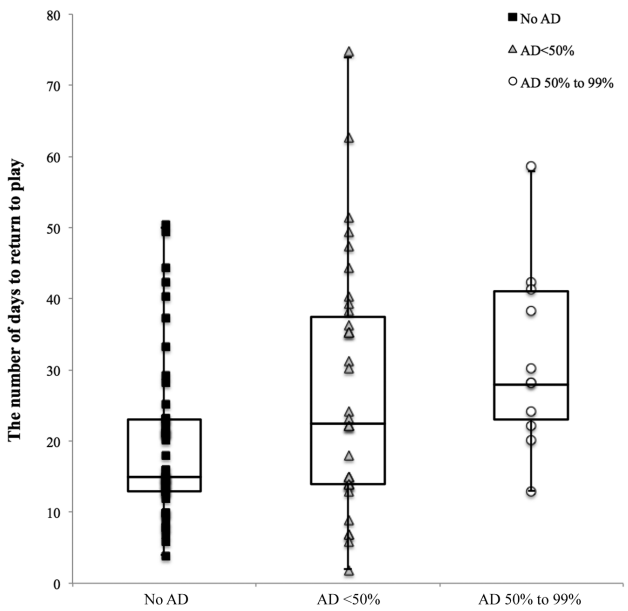

Figure 6. Box-and-whisker and scatter plots for the time to return to play with reference to the presence and severity of aponeurotic disruption. Values are presented as median, interquartile range, and range. AD, aponeurotic disruption.

arrangement.[6,24,26,34,54] Aponeuroses are a critical element of the connective tissue network and have a specialized role during dynamic activities.[23,32,35] Any AD may threaten calf muscle–tendon unit integrity and impede tissue dynamics,[45,46,48] and thus more extensive periods of rehabilitation may be required before optimal function is restored. The calf muscle–tendon unit undergoes an active stretch-shortening cycle during dynamic activities, and the interface between the muscle fibers and aponeurotic tissue (where CMSI often manifest) can be expected to experience large and rapid tensile loads on a repetitive basis. For example, it has been demonstrated that the calf muscle–tendon unit generates peak forces of 3-times body weight during jumping, 5.2-times body weight during submaximal vertical hopping, and up to 12.5-times body weight during running at 6 m/s.[29] Given the enormity of these loads and the critical role of the aponeuroses for optimal muscle–tendon unit function, it is logical that soleus injuries with AD were associated with a longer time to return to full training and RTP.

A greater severity of AD (ie, mild vs severe) was not associated with a longer RTP period in the multivariable regression. There is mixed evidence about how the severity of AD affects recovery following after muscle strain injuries.[8,41,43,44,52,54] CMSI with more severe AD have been previously reported to have a worse prognosis

TABLE 3
Clinical and MRI Data Associated With Time to Achieve Running at >90% of Maximum Speed, Return to Full Training, and Return to Play After CMSI[a]

	Running at >90% of Maximum Speed				Return to Full Training				Return to Play			
	Univariate		Multivariable		Univariate		Multivariable		Univariate		Multivariable	
	HR (95% CI)	*P* Value	AHR (95% CI)	*P* Value	HR (95% CI)	*P* Value	AHR (95% CI)	*P* Value	HR (95% CI)	*P* Value	AHR (95% CI)	*P* Value
All CMSI												
Age, y	0.96 (0.91-1.01)	.13	1.01 (0.9-1.1)	.82	0.94 (0.9-0.99)	.024[b]	0.98 (0.9-1.03)	.45	0.95 (0.9-1.0)	.05	0.99 (0.9-1.04)	.73
Dominant leg injured	0.7 (0.4-0.97)	.04[b]	0.74 (0.5-1.1)	.15	0.6 (0.4-0.9)	.02[b]	0.68 (0.5-1.03)	.07	0.75 (0.5-1.1)	.15	0.75 (0.5-1.1)	.17
Running mechanism of injury	0.5 (0.3-0.7)	.001[b]	0.57 (0.4-0.9)	.012[b]	0.52 (0.35-0.8)	.002[b]	0.66 (0.4-1.01)	.06	0.52 (0.4-0.8)	.002[b]	0.59 (0.4-0.9)	.02[b]
Primary muscle injured: soleus	0.6 (0.3-0.96)	.03[b]	0.59 (0.4-0.98)	.042[b]	0.6 (0.4-1.01)	.06	0.62 (0.4-1.03)	.07	0.59 (0.4-0.98)	.044[b]	0.62 (0.4-1.04)	.07
Aponeurotic disruption												
Present: yes/no	0.63 (0.4-0.9)	.019[b]	0.73 (0.5-1.1)	.13	0.61 (0.4-0.9)	.013[b]	0.69 (0.5-1.1)	.08	0.63 (0.4-0.9)	.02[b]	0.72 (0.5-1.1)	.11
Severity												
<50%	0.65 (0.4-0.95)	.04[b]	0.74 (0.5-1.1)	.17	0.67 (0.4-1.02)	.064	0.73 (0.5-1.1)	.16	0.67 (0.4-1.01)	.06	0.74 (0.5-1.1)	.17
50%-99%	0.6 (0.3-1.1)	.08	0.65 (0.3-1.4)	.26	0.47 (0.2-0.9)	.023[b]	0.54 (0.3-1.1)	.11	0.52 (0.3-0.99)	.04[b]	0.61 (0.3-1.3)	.19
Soleus												
Age, y	0.96 (0.9-1.02)	.19	1.01 (0.9-1.1)	.66	0.94 (0.9-0.99)	.04[b]	0.98 (0.9-1.04)	.59	0.95 (0.9-1.01)	.08	1.0 (0.95-1.1)	.99
Dominant leg injured	0.72 (0.5-1.1)	.14	0.82 (0.5-1.3)	.37	0.64 (0.4-0.99)	.04[b]	0.73 (0.5-1.2)	.18	0.79 (0.5-1.2)	.33	0.82 (0.5-1.3)	.43
Running mechanism of injury	0.43 (0.3-0.7)	<.001[b]	0.46 (0.3-0.7)	.001[b]	0.48 (0.3-0.8)	.001[b]	0.56 (0.4-0.9)	.014[b]	0.49 (0.3-0.8)	.002[b]	0.55 (0.3-0.9)	.01[b]
Aponeurotic disruption												
Present: yes/no	0.74 (0.5-1.1)	.16	0.85 (0.6-1.3)	.48	0.56 (0.4-0.9)	.008[b]	0.59 (0.4-0.9)	.018[b]	0.61 (0.4-0.9)	.026[b]	0.6 (0.4-0.9)	.025[b]
Severity												
<50%	0.7 (0.5-1.2)	.20	0.8 (0.5-1.3)	.40	0.65 (0.4-1.04)	.09	0.7 (0.4-1.1)	.08	0.65 (0.4-1.05)	.11	0.63 (0.4-1.1)	.10
50%-99%	0.8 (0.4-1.7)	.59	1.2 (0.6-2.6)	.62	0.62 (0.3-1.3)	.19	0.83 (0.4-1.7)	.60	0.73 (0.4-1.5)	.31	0.9 (0.5-2.02)	.89

[a]Referents: dominant leg injured (dominant vs nondominant), running mechanism of injury (running related vs non–running related), primary muscle injured (soleus vs other muscle), aponeurotic disruption—yes/no (AD vs no AD), aponeurotic disruption—severity (AD <50% vs no AD, AD 50% to 99% vs no AD). For functional progression outcome measures, an HR <1 indicates increased risk of worse prognosis. AD, aponeurotic disruption; AHR, adjusted hazard ratio; CMSI, calf muscle strain injuries; HR, hazard ratio; MRI, magnetic resonance imaging.

[b]Statistically significant association with outcome ($P < .05$).

TABLE 4
Clinical and MRI Data Associated With Recurrent CMSI[a]

	Early (n = 19)				Overall (n = 35)			
	Univariate		Multivariable		Univariate		Multivariable	
	HR (95% CI)	*P* Value	AHR (95% CI)	*P* Value	HR (95% CI)	*P* Value	AHR (95% CI)	*P* Value
All CMSI								
Age, y	1.3 (1.1-1.4)	<.001[b]	1.3 (1.1-1.5)	.001[b]	1.2 (1.1-1.3)	<.001[b]	1.1 (1.03-1.3)	.013[b]
Previous CMSI	5.9 (1.4-25.7)	.017[b]	4.2 (0.9-18.6)	.052	8.03 (2.5-26.2)	.001[b]	6.7 (2.03-21.9)	.002[b]
Previous ankle injury	3.9 (1.1-13.5)	.03[b]	3.9 (1.1-13.8)	.032[b]	1.4 (0.7-2.9)	.32	1.3 (0.7-2.7)	.41
Running mechanism of injury	1.4 (0.6-3.4)	.49	0.84 (0.3-2.1)	.72	2.5 (1.3-5.1)	.009[b]	1.5 (0.7-2.9)	.29
Aponeurotic disruption								
Present: yes/no	1.6 (0.6-3.9)	.31	1.1 (0.4-2.9)	.81	2.1 (1.1-4.2)	.032[b]	1.7 (0.8-3.4)	.15
Severity								
<50%	1.6 (0.6-4.2)	.29	1.1 (0.4-2.9)	.79	1.9 (0.9-3.9)	.12	1.6 (0.8-3.4)	.20
50%-99%	0.7 (0.1-5.6)	.74	0.49 (0.1-4.3)	.53	1.7 (0.6-5.2)	.36	1.2 (0.4-3.9)	.77
Soleus								
Age, y	1.2 (1.1-1.4)	.002[b]	1.2 (1.1-1.5)	.007[b]	1.2 (1.1-1.3)	.002[b]	1.1 (0.9-1.2)	.084
Previous CMSI	5.3 (1.2-23.2)	.027[b]	4.01 (0.9-17.6)	.07	7.4 (2.2-24.3)	.001[b]	5.6 (1.7-18.8)	.005[b]
Ethnicity: Indigenous Australian	0.88 (0.3-3.04)	.83	0.9 (0.4-2.6)	.90	2.2 (1.04-4.5)	.039[b]	2.2 (0.99-4.7)	.052
Running mechanism of injury	1.4 (0.5-3.6)	.51	0.79 (0.3-2.1)	.63	2.9 (1.4-6.2)	.005[b]	1.6 (0.8-3.6)	.21
Aponeurotic disruption								
Present: yes/no	1.7 (0.6-4.3)	.31	1.2 (0.4-3.4)	.76	2.2 (1.1-4.7)	.033[b]	1.9 (0.9-4.1)	.11

[a]Referents: previous CMSI (history at the time of the index injury vs no history), previous ankle injury (history at the time of the index injury vs no history), ethnicity (Indigenous Australian vs non-Indigenous), running mechanism of injury (running-related vs non–running related), primary muscle injured (soleus vs other muscle), aponeurotic disruption—yes/no (AD vs no AD), aponeurotic disruption—severity (AD <50% vs no AD, AD 50% to 99% vs no AD). For recurrence, an HR >1 indicates an increased risk of worse prognosis. AD, aponeurotic disruption; AHR, adjusted hazard ratio; CMSI, calf muscle strain injuries; HR, hazard ratio; MRI, magnetic resonance imaging.

[b]Statistically significant association with outcome ($P < .05$).

for RTP, but these results are difficult to compare with those from the current study due to differences in methods.[42,44] Given our large sample, we could adjust for the potential confounding effects of multiple variables. Previous analyses also used different approaches, such as analysis of variance,[24] or referred to the number of weeks rather than days to RTP.[25] Our findings suggest that the presence of AD may be a more relevant MRI-based injury characteristic than the precise severity of AD.

A running-related mechanism of injury had the strongest association with a longer RTP period across all CMSI. Running mechanisms were common for CMSI with AD. As discussed, the presence of AD likely adversely affects calf muscle–tendon unit function.[45,46,48] Noncontractile tissues are likely affected in all CMSI to some extent—even in cases with no visible AD on MRI.[3,5,50] At musculoaponeurotic and musculotendinous junctions, fibrils extend and interdigitate in both directions.[27] Some fibrils course throughout the muscle and form part of the supporting matrix for the contractile elements.[33,34] If MRI visualization of these structures and how they adjoin is finite, certain injury characteristics may be undetectable. This could be one reason for time loss of up to 50 days for CMSI that did not show AD on MRI. Despite the effect of pathology (ie, structural damage), contextual factors likely affect time to RTP and should be considered when estimating prognosis after any muscle strain injury, such as the demands of the sport, playing position, stage of the season, and psychology.[1]

Recurrence after CMSI was the other primary outcome that we explored. Recurrent CMSI are important to investigate since they often require an extended time to RTP.[14,18,38]

Some characteristics of recurrent CMSI in this study were consistent with hamstring muscle strain injuries: (1) the median time to recurrence was 58 days for CMSI versus 60 days for hamstring muscle strain injuries (ie, recurrences often occur early), and (2) early recurrences often involve the same muscle, gross position, and anatomical location (63.2%) (Figure 4A).[55] The broader evidence base investigating recurrent muscle strain injuries also supports our result that MRI findings of the index injury are not associated with recurrence.[21,30,50,51,53]

Clinical data about the athlete's age and injury history (early recurrence: previous ankle injury; overall: previous CMSI) were associated with a greater risk of recurrent CMSI. Recent studies similarly found older age and history of CMSI to be risk factors for CMSI,[19,39] and clinical data provide better information than MRI findings about the risk of recurrent hamstring muscle strain injuries.[12] A previous muscle strain injury can result in persistent strength deficits,[6,17,38] atrophy,[6,25,38,49] and noncontractile tissue infiltration,[20,38] as well as neurophysiological changes (eg, inhibition).[17,38] If such maladaptive changes occur, it could negatively affect muscle function and increase the likelihood of a recurrence. Future research is needed to understand the extent of any maladaptive changes after CMSI and the most effective strategies for reducing the risk of recurrent CMSI.

Strengths and Limitations

This study is the largest evaluation of the potential association of clinical and MRI data with RTP and recurrence after CMSI. Findings were generated from a homogeneous data set, sourced from a single elite competition which has consistent scheduling of games and training activities, contributing to the strengths of the conclusions in this population. Radiologists were also blinded to clinical and outcome data when evaluating the MRI scans, reducing the likelihood of bias. This study has several notable limitations that should be acknowledged as well. Data were unable to be compared with the entire AFL cohort because data relating to uninjured players were not reported to the Soft Tissue injury Registry of the AFL. Club medical personnel were unable to be blinded to MRI findings when managing the injured players, which could have influenced prognosis. There is also some potential overlap between similar mechanisms of injury; to overcome this problem, we grouped similar activities in the regression analyses (ie, running-related mechanism vs non–running related mechanism). Further analysis in a larger cohort is required to determine whether the clinical and MRI data identified to be associated with RTP and recurrence are predictors of these outcomes. Findings from a single sport may lack generalizability to outcomes after CMSI in other populations as well.

CONCLUSION

The anatomical location of injury was not associated with the time to RTP after CMSI, but MRI evidence of AD and information about the mechanism of injury may inform RTP prognosis. Clinical data are most salient to understand the risk of recurrent CMSI.

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